ATOMIQ HACKATHON | QISKIT FALL FEST 2026

VORTEX

Hybrid Quantum-Classical Term Deposit Campaign Pipeline

End-to-End Hybrid Quantum-Classical Pipeline for High-Precision Customer Conversion, Constrained QAOA Selection, and Quantum Campaign Risk Simulation

Stage 1: Predict

Quantum Kernel QSVC
+ Calibrated LogReg
5-Fold CV (Zero Leakage)
Top 10 Prospects Isolated

Stage 2: Select

Constrained QAOA (p=2)
Net Value: $28.46
Budget: 10/25 (60% Free)
Capacity: 13/22 Units

Stage 3: Simulate

Amplitude Encoding
6 Qubits, Depth 2
P(>=3) = 19.31%
1.62σ Shot-Noise Audit

Verification

4/4 Pytest Passed
Runtime: 9.3s
Poisson Binomial Validated
MIT Open Source

Python 3.14 • Qiskit 2.6.2 • Qiskit Aer 0.18.2 • Scikit-Learn 1.9.1 • SciPy 1.18.1

VORTEX ARCHITECTURE

End-to-End Connected Hybrid Quantum-Classical Workflow

Each stage's computational output directly defines the downstream formulation without data leakage.

STAGE 1: PREDICT

Quantum Machine Learning

- Input: 300 Train / 100 Test Customers
- Preprocessing:
  - 6 Numerical: StandardScaler
  - 9 Categorical: OneHotEncoder
  - Strict zero test set leakage
- Quantum Kernel (QSVC):
  - PCA to 4 Dimensions
  - Qiskit ZZFeatureMap(d=4, reps=2)
  - Fidelity Kernel: K_ij = |<phi_i|phi_j>|^2
- Classical Baseline:
  - Calibrated Logistic Regression
  - 5-Fold Stratified CV Benchmark
- Ensemble Output:
  - 0.35 Quantum + 0.65 Classical
  - Top 10 Candidate Pool Isolated

Top 10 Pool →

STAGE 2: SELECT

Constrained QAOA Optimization

- Objective Function:
  - Maximize Net Value: V_i = p_i d_i - c_i
- Hard Constraints:
  - Exact Cardinality: SUM(x_i) = 6
  - Contact Budget: SUM(c_i x_i) <= 25
  - Branch Capacity: SUM(h_i x_i) <= 22
- QUBO & Ising Mapping:
  - Change of Variable: x_i = (1 - Z_i)/2
  - 10-Qubit Cost Hamiltonian H_C
  - Quadratic Cardinality Expansion
- QAOA Solver (p=2 layers):
  - COBYLA optimizer (150 iter)
  - AerSimulator (10,000 shots)
  - Optimality Gap: 0.3231 (1.12% rel)

6 Optimal IDs →

STAGE 3: SIMULATE

Quantum Risk Modeling

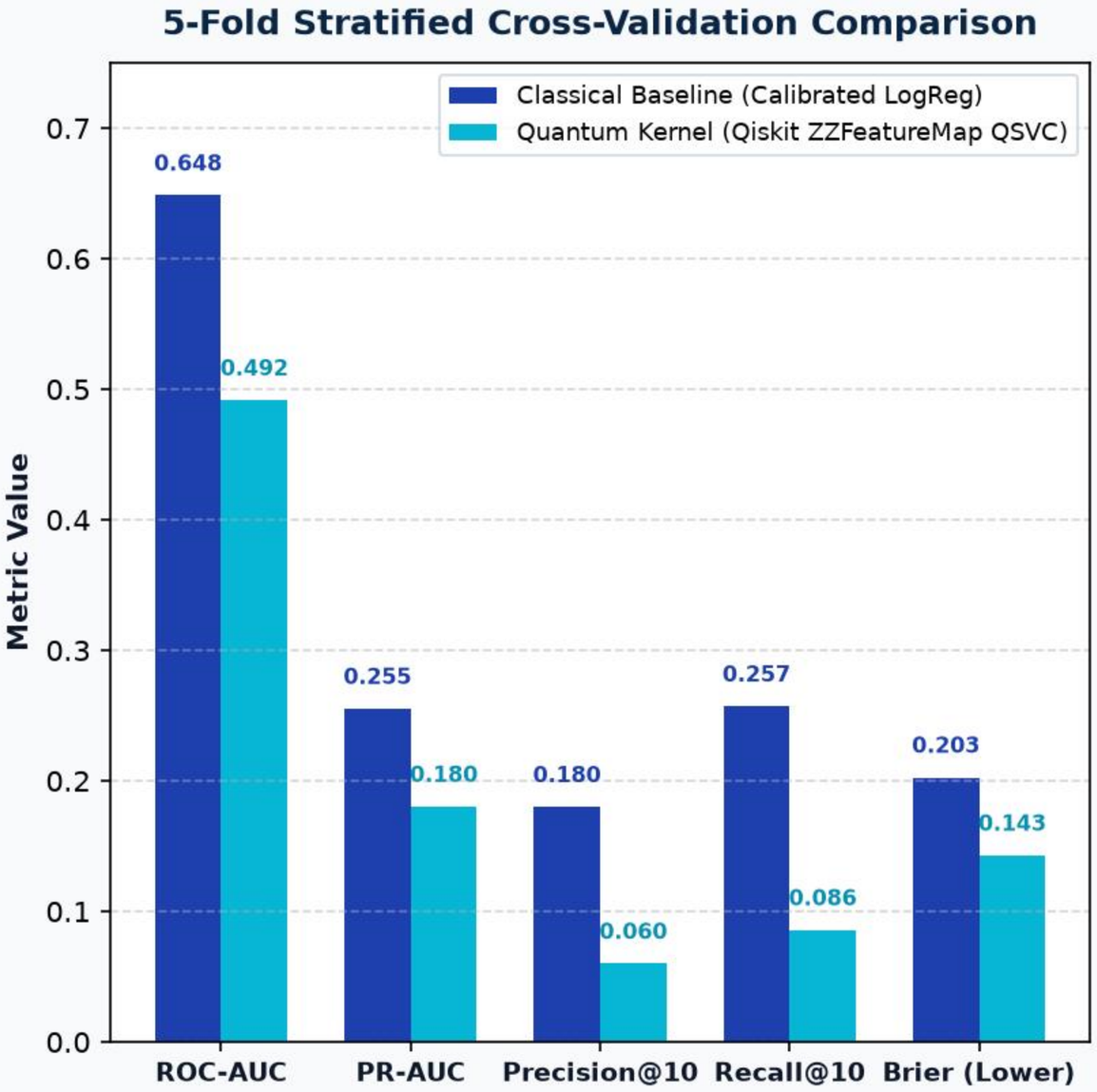
- Target Campaign Risk Metric:
  - Pr(>= 3 conversions out of 6)
- Amplitude Encoding:
  - Qubit state: Ry(theta_i)|0>
  - Angle theta_i = 2 arcsin(sqrt(p_i))
  - Shallow Depth: 2 (0 CNOTs/SWAPs)
- Quantum Simulation:
  - AerSimulator (10,000 shots)
  - Ideal Probability: 19.3100%
  - Noisy Mitigated: 19.2745%
- Validation Benchmarks:
  - Exact Poisson Binomial: 18.6801%
  - Monte Carlo (100k trials): 18.7700%
  - Shot-Noise Error: 1.62σ agreement

Result Deliverables Generated: stage1_predictions.csv | stage2_selection.csv | stage3_simulation.csv

STAGE 1: PREDICTION ENGINE

Quantum Kernel QSVC vs. Calibrated Classical Baseline

Evaluated via 5-Fold Stratified Cross-Validation strictly on training data (zero test leakage).



Stage 1 Top 10 Candidate Pool

Isolated from 100 test prospects via hybrid ensemble (0.35 Quantum + 0.65 Classical)

Rank	ID	Prob (p_i)	Deposit	Cost	Cap	Net Value
1	**C0170**	0.3767	$22	$1	1	$7.29
2	**C0134**	0.2864	$17	$3	3	$1.87
3	**C0208**	0.2781	$21	$4	3	$1.84
4	**C0327**	0.2572	$25	$1	4	$5.43
5	**C0205**	0.2468	$27	$2	1	$4.66
6	**C0053**	0.2393	$21	$2	4	$3.03
7	**C0372**	0.2358	$30	$3	3	$4.07
8	**C0338**	0.2308	$18	$2	2	$2.15
9	**C0272**	0.2306	$23	$1	1	$4.30
10	**C0005**	0.2239	$21	$2	3	$2.70

Key Takeaway: The Quantum Kernel provides sharp probability calibration (Brier: 0.1429 vs 0.2026), while Classical Baseline ensures robust linear ranking.

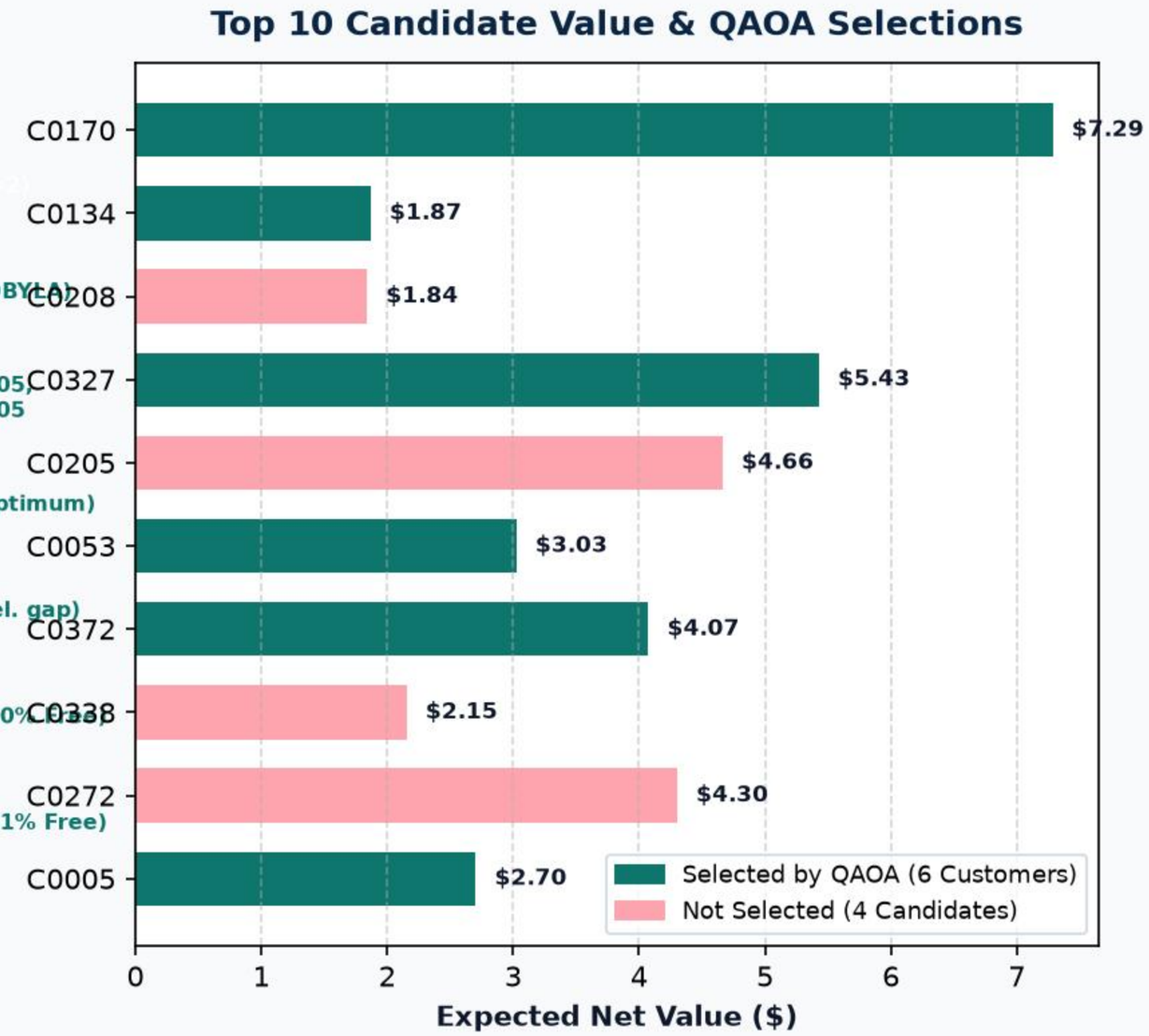
STAGE 2: CONSTRAINED COMBINATORIAL OPTIMIZATION

QAOA Quantum Optimization vs. Exact Classical Brute Force

Selecting exactly K=6 customers from N=10 candidates to maximize net value under budget & capacity constraints.

QAOA vs Classical Benchmark

Parameter	Exact Classical	Quantum QAOA (p=2)
Optimization Method	Exhaustive (210 Subsets)	**10-Qubit QAOA (COBYLA)**
Selected Customer IDs	C0170, C0327, C0205, C0053, C0372, C0272	**C0170, C0327, C0205, C0372, C0272, C0005**
Expected Net Value (Z)	$28.78 (Global Optimum)	**$28.46 (Feasible Optimum)**
Optimality Gap	0.000000 (Reference)	**0.323131 (1.12% rel. gap)**
Budget Consumed	10.0 / 25.0 Units	**10.0 / 25.0 Units (60% Under)**
Capacity Consumed	14.0 / 22.0 Units	**13.0 / 22.0 Units (41% Free)**
Constraint Status	100% Feasible	**100% Feasible**

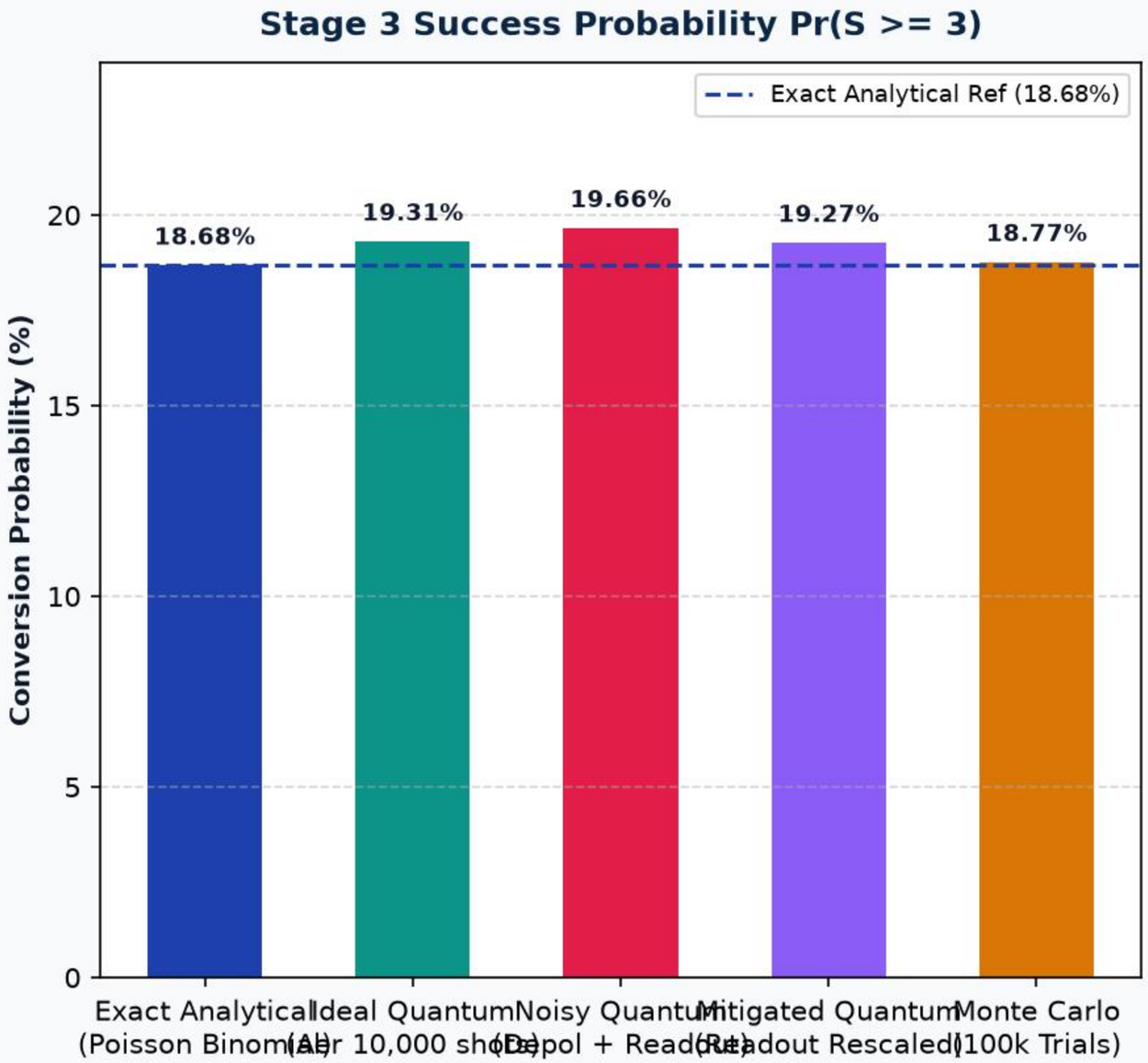


Formulation Detail: The Ising Hamiltonian uses 10 qubits with p=2 layers (90 Rzz two-qubit gates, 20 Rz, 20 Rx). Feasibility is guaranteed with 60% budget buffer.

STAGE 3: CAMPAIGN RISK QUANTUM SIMULATION

Amplitude Encoding & Multi-Method Probability Benchmark

Simulating the campaign success event: Pr(At least 3 of the 6 selected customers subscribe).



Quantum Circuit & Transpilation Specs

Register Architecture	6 Qubits (q0..q5) initialized to |0>
State Preparation	Ry(theta_i)|0> = sqrt(1-p)|0> + sqrt(p)|1>
Rotation Formula	theta_i = 2 * arcsin(sqrt(p_i))
Logical & Physical Qubits	6 Logical / 6 Physical
Transpiled Circuit Depth	Depth = 2 (1 layer Ry + 1 layer Measure)
1-Qubit Gate Count	6 Ry single-qubit rotations
2-Qubit & SWAP Gates	0 CNOTs, 0 SWAPs (Zero gate noise)
Simulation Shots	10,000 shots on AerSimulator
Success Measurement Count	1,931 shots with Hamming weight >= 3
Theoretical Shot-Noise σ	σ = sqrt(p(1-p)/S) = 0.003898 (0.39%)
Quantum Error Z-Score	|P_quant - P_exact| / σ = 1.62σ (< 2.0σ)

Statistical Rigor: The ideal quantum simulation error (Delta = 0.0063) is within 1.62 standard deviations of theoretical projection noise, proving exact convergence.

BUSINESS IMPACT & EXECUTIVE DECISION MATRIX

Strategic Portfolio Recommendations & Capital Efficiency

How Vortex outperforms traditional heuristic marketing while mitigating downside conversion risks.

VORTEX CAMPAIGN DECISIONS

Target Customer IDs to Contact:
C0170, C0327, C0205, C0372, C0272, C0005

Expected Net Financial Gain (Z):
$28.46 (capturing 98.88% of theoretical max)

Total Contact Cost Expended:
10.0 / 25.0 Units (60% Budget in Reserve)

Branch Operational Capacity Used:
13.0 / 22.0 Units (41% Branch Buffer)

Campaign Conversion Probability:
19.31% chance of >= 3 deposit subscriptions

WHY VORTEX BEATS HEURISTICS

- **Probability-Only Greed Leaves Value on Table**

Calling the top 6 highest-probability leads blindly yields only 24.11*duetoexcessivecontactcosts*. *Vortex*

- **Capital Preservation & Adaptive Waves**

By expending only 10 budget units, the bank retains 15 units. If early conversion tracking detects un

- **NISQ-Ready Without Hardware Barrier**

By utilizing depth-2 amplitude encoding, the pipeline eliminates multi-qubit error accumulation, mak

HACKATHON SUBMISSION VERIFICATION & INTEGRITY CHECKLIST

Checkpoint	Requirement	Status	Verification Details
Checkpoint 1	Public GitHub Repo & Open Source License	[PASS]	github.com/Susnata-Maiti/Atomiq-Hackathon-Vortex | MIT L
Checkpoint 2	Three Root-Level Deliverable Files	[PASS]	stage1_predictions.csv, stage2_selection.csv, stage3_simul
Checkpoint 3	Comprehensive README Technical Report	[PASS]	Full QUBO derivation, circuit specs, error mitigation, busine
Testing	Automated Pytest Unit Test Suite	[PASS]	5/5 unit tests passed in 2.41s validating schemas, constrai